

DELTA TRACE: EFFICIENT SIGNED ATTRIBUTION FOR REASONING LANGUAGE MODELS

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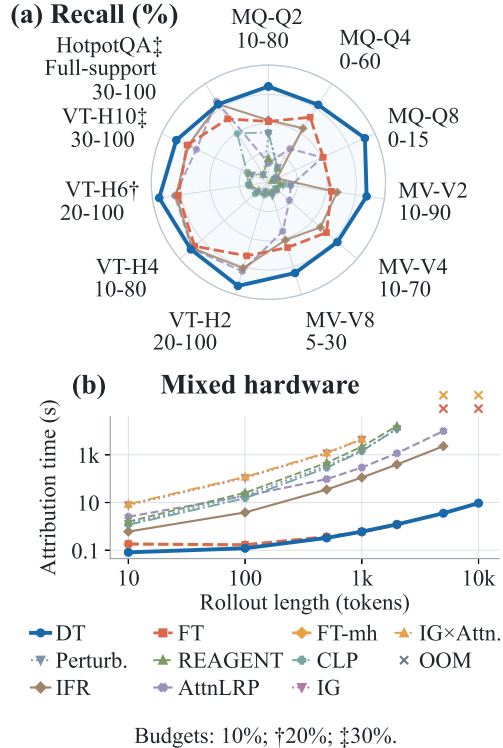
ABSTRACT

We introduce DELTA TRACE, which assigns signed scores to input words to explain a fixed, complete reasoning response. It compares the response’s log-probability under the original prompt and a reference that replaces source words with end-of-sequence tokens. Analytic rules convert the observed activation changes into coefficients for one reverse pass, analogous to gradient backpropagation. The token scores sum to the response-score difference. The rules follow content and control together. Attention traces values and their selection, while gated delta memory traces stored content and the keys, queries, and gates controlling retention, writing, and reading. Analytic normalization accounts for competition among attention sources. FlashAttention-style tiling and Flash Linear Attention chunks give sequence-linear auxiliary storage at fixed model dimensions and chunk size. On Qwen3-8B retrieval and reasoning tasks, DELTA TRACE achieves the lowest deletion-faithfulness scores (RISE and MAS) on most tasks and the highest needle-in-a-haystack Recall, exceeding FlashTrace by 5.67–24.33 percentage points. On Qwen3.5-9B, across 13 tasks and 520 examples, DELTA TRACE lowers RISE on 10 tasks and MAS on 12 relative to FlashTrace. Batching with GPU checkpoints increases throughput by 35.3%.

1 INTRODUCTION

Input attribution assigns scores to the words in a prompt to explain a model’s response. A reasoning response depends on both the information available and the computations that select and combine it (Ferrando & Voita, 2024). Figure 1(d) shows a passage containing a playwright, William Shakespeare, and a composer, William Walton. The names supply candidate answers, and the phrase “the author of the play” specifies which role to use. An explanation should follow both the supplied content and the control that directs its use. It should also distinguish words that support the response from words that oppose it.

We ask how replacing selected prompt tokens would change the model’s score for a chosen response span. The reference replaces these source tokens with end-of-sequence (EOS) tokens at the same positions. Both executions score the same target span using the fixed response prefix. The target can be an answer span or the complete reasoning and answer. We assign each source token a signed share of the resulting log-probability difference, so the contributions add across words or passages.



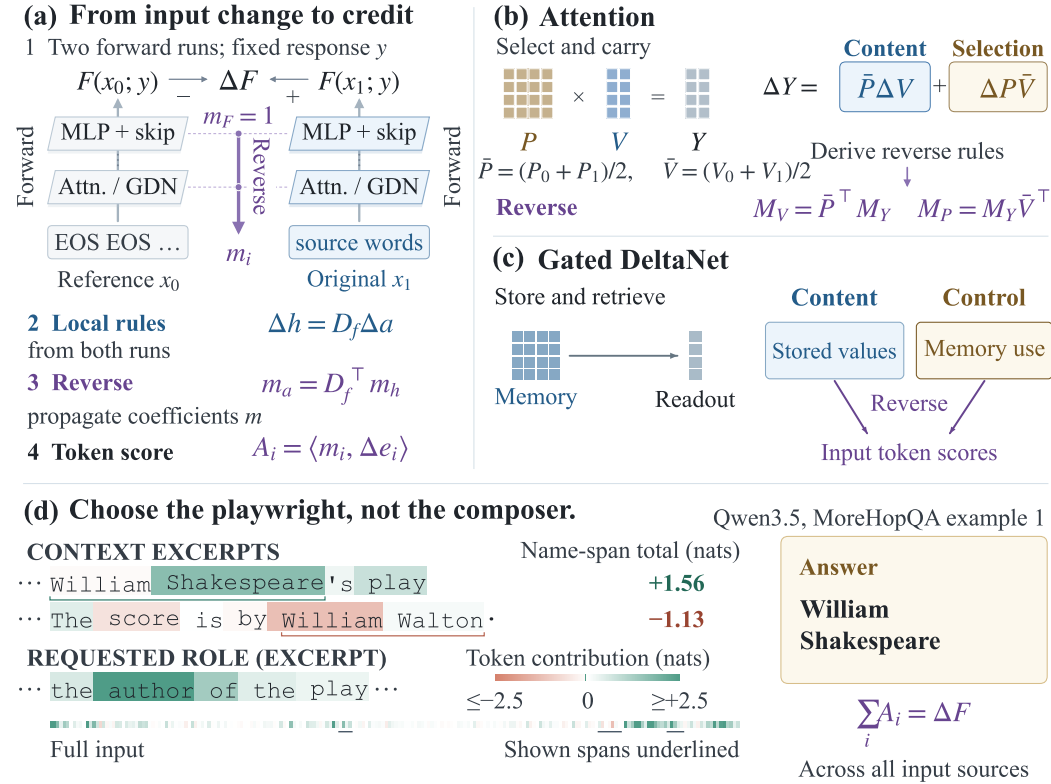


Figure 1: **DeltaTrace follows content and control.** (a) Gray arrows show the original and reference forward runs. The purple reverse pass propagates coefficients backward to compute token scores that sum to their score difference ΔF . (b) Attention assigns credit through values and selection. Bars denote averages of the two runs. (c) GDN assigns credit through stored values and the operations controlling their use. Both paths continue to input token scores. (d) Qwen3.5 assigns positive scores to the playwright’s name and negative scores to the composer’s. Colors saturate at ± 2.5 nats, and annotations sum the marked names. Ellipses omit text. The target includes the complete stored response and EOS.

Attribution must specify both the score being explained and how it is assigned to input sources. Integrated Gradients and DeepLIFT assign prediction differences relative to a reference (Sundararajan et al., 2017; Shrikumar et al., 2017). Relevance propagation and representation decomposition trace information through attention and recurrent states (Ali et al., 2022; Achibat et al., 2024; Rezaei Jafari et al., 2024; Modarressi et al., 2023; Pitorro & Treviso, 2025). FlashTrace follows source importance through multi-token responses (Pan et al., 2026). For the fixed-response question above, the target is a finite log-probability difference. Assigning that difference to source tokens requires following changes in both their content and the operations that select, store, and retrieve it.

The difficulty is that content and control change together. Changing a token can alter both its value vector and the attention weights that select it. Those weights compete because their sum is one. In gated memory, a new value changes a state assembled from earlier tokens, while keys, queries, and gates determine what is written, retained, and read (Yang et al., 2025). Tracing only the transported content leaves these selection and memory decisions outside the explanation. Tracing both requires local rules that account for their joint changes and can be composed across the whole response.

DELTATRACE uses two forward executions to obtain the original and reference activations and their differences. Analytic rules relate each operation’s output change to its input changes, following content and control together. A single reverse pass follows the structure of gradient backpropagation, using these rules in place of local derivatives. Starting from the response score, it computes

coefficient vectors for the input embeddings. Their dot products with the corresponding embedding differences give signed token scores that sum to the response-score difference.

Contributions.

- **Attribution for complete responses.** DELTATRACE assigns signed token contributions to the log-probability difference between original and reference prompts for a fixed, complete response. Two forward executions and one finite reverse pass produce scores that sum to this difference.
- **Content and control.** Analytic finite rules for attention and Gated DeltaNet trace values together with the operations that select, retain, write, and read them.
- **Efficient computation.** Finite reverse propagation uses FlashAttention-style tiles (Dao et al., 2022) and Flash Linear Attention (FLA) chunks (Yang & Zhang, 2024). Auxiliary storage grows linearly with sequence length at fixed model dimensions and chunk size.

On Qwen3-8B retrieval and reasoning tasks, DELTATRACE achieves the lowest deletion-faithfulness scores on most tasks and the highest evidence recovery across the six needle-in-a-haystack tasks (Section 4). Qwen3.5-9B measurements evaluate finite reverse propagation through attention and gated memory. Appendix F discusses the connection between counterfactual return differences and policy advantages (Sutton et al., 1999; Foerster et al., 2018).

2 DELTATRACE

2.1 THE COUNTERFACTUAL QUESTION AND ATTRIBUTION TARGET

We explain how replacing selected prompt tokens changes the score of a fixed target span. The inputs are a trained language model with fixed parameters θ , a prompt x containing the context and question, and a stored response $y = (y_1, \dots, y_L)$ of L tokens. Let $\mathcal{T}$ be the positions of the target span within y . Both executions score these positions using the same preceding response tokens. The scalar target score is

$$F(x; y) = \sum_{t \in \mathcal{T}} \log p_{\theta}(y_t \mid x, y_{<t}), \quad (1)$$

where p_{θ} is the model’s next-token probability and $y_{<t}$ is the response prefix before position t . A higher score means the model assigns greater probability to the target span given its preceding tokens. For full-response attribution, y includes the reasoning, answer, and end-of-sequence (EOS) token, and $\mathcal{T} = \{1, \dots, L\}$. The response tokens and target positions stay fixed, while their hidden representations can change as they read the different prompts.

The prompt positions chosen for explanation are the source positions. They can include context words and question words. Let x_1 be the original prompt and x_0 its reference. We construct x_0 by replacing all source tokens together with EOS at the same positions, preserving token alignment between the two executions. The remaining prompt tokens stay fixed. This reference text sets the zero point for attribution. The reference execution processes the full sequence, including the replacement tokens, to score the same target span. The two forward executions give $\Delta F = F(x_1; y) - F(x_0; y)$. A positive ΔF means these replacements lower the target score. We assign each source token a signed share A_i of this joint score difference, with the contributions adding to ΔF . Positive and negative token scores express support and opposition under this reference, as illustrated in Figure 1.

Subscripts 0 and 1 denote reference and original activations, and $\Delta U = U_1 - U_0$ denotes their difference. For source token i , $e_{i,0}$ is the EOS embedding and $e_{i,1}$ is its original embedding, so $\Delta e_i = e_{i,1} - e_{i,0}$. Tokens held fixed have zero embedding difference. The reverse computation below converts these vector differences into the scalar scores A_i .

2.2 TWO FORWARD RUNS AND ONE REVERSE PASS

Figure 1(a) shows four steps. We collect paired activations in two forward executions, obtain local rules from those activations, propagate coefficients backward, and form token scores at the input.

The forward executions supply the actual changes, and the reverse pass computes the coefficients that turn those changes into contributions to ΔF .

For an operation $h = f(a)$, a collects its input activations and h is its output. Let J_f be the matrix of local derivatives at one execution. A Jacobian–vector product (JVP) maps an infinitesimal input change da to an output change dh . A vector–Jacobian product (VJP) maps an incoming gradient g_h back to the input gradient g_a .¹ Table 1 shows the corresponding finite relations.

Table 1: **From gradients to finite reverse propagation.** Paired activations define the finite map D_f that replaces J_f . Changes appear in the left column, and coefficients propagate backward in the right column.

Method	Change relation (JVP form)	Backward coefficients (VJP form)
Gradient backpropagation	$dh = J_f da$	$g_a = J_f^\top g_h$
DELTA TRACE	$\Delta h = D_f \Delta a$	$m_a = D_f^\top m_h$

An analytic rule constructs D_f from the two executions so that $\Delta h = D_f \Delta a$. This identity specifies how the input changes account for the output change. For a scalar nonlinearity, $D_f = \Delta h / \Delta a$ is the secant slope joining the observed points, replacing the tangent slope $f'(a_1)$. At coincident inputs, it becomes the derivative. For operations with multiple inputs, the rules below specify their joint allocation. The algorithm obtains its activation differences from the two forward executions and evaluates the transpose products during the reverse pass.

The coefficient vector m_h supplies a multiplier for each component of Δh . Their dot product $\langle m_h, \Delta h \rangle$ is the contribution assigned at that output. Substituting the local change relation gives $\langle m_h, D_f \Delta a \rangle = \langle D_f^\top m_h, \Delta a \rangle$. Reading off the input coefficients yields the reverse rule $m_a = D_f^\top m_h$.

Starting with $\Delta F = 1 \cdot \Delta F$, the reverse pass repeatedly rewrites weighted output changes as weighted input changes. At a residual addition, the coefficient passes to each summand. Coefficients returning from multiple consumers add. This reaches the input embeddings and gives

$$A_i = m_i^\top \Delta e_i, \quad \sum_i A_i = \Delta F. \quad (2)$$

Each component of m_i multiplies the corresponding embedding change, and summing those products gives the token attribution score A_i . A fixed token has $\Delta e_i = 0$ and contributes zero. The scores use natural-log units (nats) and add into name, sentence, or passage scores. Appendix A.1 derives the conservation identity.

2.3 WHAT ATTENTION CARRIES AND SELECTS

An attention head lets each position read its own representation and those of earlier positions. Its query describes what to retrieve, each source key describes how to match that source, and each value carries the source’s content. Query–key similarity scores are normalized by softmax into attention weights. Let P_{ij} be the weight with which position i reads source j , and let the rows of V contain the source value vectors. The output $Y = PV$ is their weighted sum.

Changing a source can change both its value and the weight assigned to it. To separate their effects in PV , we consider two algebraic decompositions, with V changing before P or P before V . The first order weights ΔV by P_0 and applies ΔP to V_1 . The second uses P_1 and V_0 , respectively. We average the two decompositions to give both orders equal weight. Writing $\bar{P} = (P_0 + P_1)/2$ and $\bar{V} = (V_0 + V_1)/2$, we obtain

$$\Delta Y = \underbrace{\bar{P} \Delta V}_{\text{content change}} + \underbrace{\Delta P \bar{V}}_{\text{selection change}}. \quad (3)$$

The two terms sum to the observed output difference, with the interaction $\Delta P \Delta V$ shared equally between them. In the running example, the name supplies content, while the requested role can change which person’s information is selected.

¹PyTorch documentation for `torch.func.jvp` and `torch.func.vjp`.

The two forward executions supply the endpoints for $\bar{P}$ and $\bar{V}$. Given an incoming coefficient matrix M_Y , Equation 3 rewrites $\langle M_Y, \Delta Y \rangle$ as $\langle \bar{P}^\top M_Y, \Delta V \rangle + \langle M_Y \bar{V}^\top, \Delta P \rangle$. The coefficients of the two input changes are therefore

$$M_V = \bar{P}^\top M_Y, \quad M_P = M_Y \bar{V}^\top. \quad (4)$$

These updates use $\bar{P}$, $\bar{V}$, and the incoming M_Y directly. The identity for ΔY explains their allocation. The quantities returned backward are M_V and M_P . Both coefficient paths continue to the input representations that produced V and P .

Selection is relative because the attention weights in a row sum to one, so increasing one source’s share reduces the total share of the others. The finite softmax rule carries this competition backward by subtracting a weighted row average from the incoming coefficients. Its weights depend on the probabilities in both executions. A source therefore receives selection credit relative to the alternatives that the same query could read. The coefficients then pass through the query–key scores to their input representations. The same construction handles the vocabulary normalization used to score each response token. Appendix A.2 gives the rule and its derivation.

2.4 WHAT MEMORY KEEPS AND UPDATES

Qwen3.5 interleaves attention with Gated DeltaNet memory (Yang et al., 2025). A memory layer carries a state matrix S_{t-1} from one token position to the next. This matrix stores associations between keys and values. At position t , the model supplies a key k_t , a value v_t , a read query q_t , a retention gate α_t , and a write gate β_t . The two gates are scalars controlling how much information is kept and written.

The update first retains $T_t = \alpha_t S_{t-1}$. Reading this retained state at key k_t gives $T_t^\top k_t$, the value already stored for that key. The correction $u_t = \beta_t(v_t - T_t^\top k_t)$ writes the difference between the incoming and stored values. The new state and the output read by the query are

$$S_t = \underbrace{T_t}_{\text{retained content}} + \underbrace{k_t u_t^\top}_{\text{write correction}}, \quad o_t = S_t^\top q_t. \quad (5)$$

Here o_t is the memory readout at position t . It is normalized to n_t , multiplied elementwise by an activated output-gate vector s_t , and linearly projected to form the layer output. This sequence lets early information survive later updates and reach the response.

Reverse coefficients first split between normalized content and the output gate. The memory readout then sends coefficients to the state and query. Reversing the updates carries them to earlier states, values, keys, and retention and write gates. The coefficients continue to the input embeddings, where Equation 2 produces token scores. A word can therefore receive credit for providing information and for changing how that information is stored or retrieved.

As in attention, the products in memory updates allow two decompositions that exchange the roles of the original and reference activations. Each choice defines a finite reverse recurrence over the complete memory sequence, and both account for the same readout changes. During the model’s single reverse pass, each GDN layer applies the two recurrences to the same readout coefficients. We average the returned coefficients for values, keys, queries, and retention and write gates before continuing backward. This gives the two executions equal roles within each layer. The output product $n_t \odot s_t$ also uses symmetric allocation between content and gate changes. Appendix A.4 gives the finite recurrence and output-gating rules.

The remaining network operations complete the same reverse pass in both model families. Linear layers use their weight matrices, additions pass coefficients to their inputs, and nonlinearities and normalizations use finite rules. Appendix A.3 lists the choices for each operator, including symmetric interaction allocation for attention values, query–key products, and feed-forward gating.

3 EFFICIENT COMPUTATION

The reverse pass needs coefficients for input activations at each layer. Keeping an attention matrix for every layer would require storage for all token pairs. DELTATRACE instead computes local transpose products in small blocks and accumulates their results directly into the input coefficients. The attention and memory paths use the following two forms of this idea.

Attention in tiles. A tile is a small block of query and source positions. Following FlashAttention (Dao et al., 2022), DELTATRACE reconstructs the two executions’ attention probabilities for one tile, uses them to update query, key, and value coefficients, and reuses the workspace for the next tile. The softmax rule needs a weighted average over the whole row. One sweep collects those row statistics. Subsequent sweeps use them when propagating through each tile. For N input-plus-response positions and head dimension d , the stored vectors and row statistics occupy $O(Nd)$ auxiliary space per head. Dense attention requires $O(N^2d)$ arithmetic.

Memory in chunks. A chunk is a consecutive group of tokens. The memory state at a chunk boundary summarizes what earlier tokens passed forward. In reverse, a coefficient for that state carries credit back to the preceding chunk. Within each chunk, matrix products distribute the coefficients across its reads, writes, and gates. This organization uses Flash Linear Attention (FLA) (Yang & Zhang, 2024). At fixed state dimensions and chunk size, the stored token coefficients and boundary states grow linearly with sequence length.

The complete attribution call includes two forward executions and one reverse traversal. It recomputes intermediate activations one layer at a time during the reverse traversal, so each layer’s workspace can be reused for the next. Appendix B details the storage accounting and accelerated operations. Section 4 measures the complete call’s time and memory.

4 EVALUATION

We evaluate whether token scores recover the evidence used in a response, predict the effect of removing that evidence, and can be computed efficiently.

Tasks and methods. Experiments use Qwen3-8B and the 1,243 examples released with FlashTrace (Pan et al., 2026). These comprise ten RULER tasks with 100 examples each, HotpotQA with 48, MATH with 100, and MoreHopQA with 95.² Needle-in-a-haystack (NIAH) tasks retrieve information among distractors. Multi-query tasks (MQ-Q2/Q4/Q8) ask for 2, 4, or 8 keys, while multi-value tasks (MV-V2/V4/V8) ask for 2, 4, or 8 values of one key. Variable Tracking (VT) follows variable assignments. Here H is the number of binding hops and C the number of parallel chains. HotpotQA and MoreHopQA require combining facts, while MATH contains mathematical reasoning problems. The complete stored reasoning and answer plus EOS is scored for faithfulness. We compare DELTATRACE (DT) with FlashTrace (FT), Perturbation, REAGENT, Context Length Probing (CLP), Information Flow Routes (IFR), and Attention-aware Layer-wise Relevance Propagation (AttnLRP). DT and FT use the same evaluation examples. VT/HotpotQA recovery compares all seven methods.

Evidence recovery. Recall is the fraction of annotated evidence recovered among the highest-ranked sources. NIAH evidence consists of the inserted key–value sentences, and VT evidence consists of the assignment statements along the target variable chain (Pan et al., 2026). A token budget limits how much input the ranking may select. For NIAH, DT attributes the complete response plus EOS. We select the top 10% of eligible input tokens ranked by positive-part attribution scores. Eligible tokens are the source positions marked for evaluation in the released examples. VT measures token Recall in the passage body, with budgets of 10%, 10%, 20%, and 30% for H2, H4, H6, and H10, and attributes the answer tokens. HotpotQA reports Full-support Recall@30%. An example scores one if every official supporting fact is selected, and zero otherwise. It attributes the full response, ranks passage sentences by the sum of their positive token contributions, and selects the longest ranking prefix within 30% of all body tokens. FT uses three recursive tracing steps (K3) for recovery. Table 4 reports these values.

Deletion faithfulness. The released deletion test replaces high-ranked source tokens with EOS in 20 steps, rescoring the same complete response after each step. The score is normalized between the full-input and fully-deleted endpoints. A good ranking causes an early score drop because it removes evidence the model uses. The RISE deletion score summarizes the area under this curve. Magnitude Aligned Scoring (MAS) also compares the remaining response score with the remaining

²<https://github.com/wbopan/flashtrace/releases/tag/table1-data-v1>

Table 2: RISE $\downarrow$ on the complete released Qwen3-8B tasks. FT is FlashTrace (Pan et al., 2026) and DT is DELTATRACE. DT uses signed ranking. FT uses the same evaluation examples as DT. n is the released task size. Bold marks the lowest observed mean across all seven methods.

Task	n	Perturbation	REAGENT	CLP	IFR	AttnLRP	FT	DT
MQ-Q2	100	0.0945	0.1174	0.0977	0.0747	0.1955	0.0682	0.0723
MQ-Q4	100	0.2389	0.2599	0.2534	0.1147	0.2628	0.1134	0.1448
MQ-Q8	100	0.4986	0.4865	0.5096	0.3709	0.3766	0.3519	0.4286
MV-V2	100	0.1343	0.1876	0.1559	0.0691	0.1403	0.0686	0.0597
MV-V4	100	0.1862	0.2107	0.2166	0.0726	0.1934	0.0704	0.0587
MV-V8	100	0.3514	0.3685	0.3929	0.2051	0.2852	0.1828	0.1159
VT-H2-C3	100	0.3844	0.4378	0.3742	0.1609	0.3191	0.1317	0.1003
VT-H4-C1	100	0.3538	0.3971	0.3277	0.1018	0.3239	0.1102	0.1013
VT-H6-C1	100	0.4584	0.4860	0.4225	0.1245	0.3378	0.1224	0.1109
VT-H10-C1	100	0.4663	0.4953	0.4510	0.1526	0.3573	0.1427	0.1403
HotpotQA	48	0.1333	0.1445	0.1013	0.0745	0.1548	0.0331	0.0275
MATH	100	0.3802	0.3878	0.3616	0.3537	0.3684	0.3487	0.2321
MoreHopQA	95	0.2489	0.2652	0.2487	0.1455	0.2857	0.1281	0.0703

Table 3: MAS $\downarrow$ on the same released tasks and seven methods as Table 2. DT uses its positive source contributions. Bold marks the lowest observed mean.

Task	n	Perturbation	REAGENT	CLP	IFR	AttnLRP	FT	DT
MQ-Q2	100	0.1442	0.1966	0.1663	0.1397	0.3261	0.1628	0.1176
MQ-Q4	100	0.3270	0.3574	0.3203	0.1770	0.4509	0.1937	0.2011
MQ-Q8	100	0.7093	0.6942	0.6566	0.4600	0.6020	0.4271	0.6047
MV-V2	100	0.1870	0.2756	0.2159	0.1343	0.2290	0.1569	0.1046
MV-V4	100	0.2436	0.2905	0.2804	0.1416	0.3248	0.1605	0.1058
MV-V8	100	0.4580	0.4938	0.5114	0.2746	0.4752	0.2477	0.1717
VT-H2-C3	100	0.5513	0.6680	0.4900	0.2309	0.5214	0.1873	0.1486
VT-H4-C1	100	0.5168	0.6031	0.4199	0.1479	0.5475	0.1656	0.1406
VT-H6-C1	100	0.6837	0.7453	0.5648	0.1727	0.5723	0.1787	0.1537
VT-H10-C1	100	0.7013	0.7405	0.5967	0.2011	0.5915	0.1983	0.1842
HotpotQA	48	0.2203	0.2351	0.1895	0.1655	0.2493	0.1276	0.1002
MATH	100	0.5202	0.5358	0.4907	0.4897	0.5244	0.4461	0.3323
MoreHopQA	95	0.3470	0.3679	0.3478	0.2277	0.4579	0.2049	0.1546

share of positive attribution (Pan et al., 2026). Lower values are better for both metrics. DT uses signed scores for RISE. For MAS, it sets negative scores to zero, ranking sources by their positive support for the response. Appendix C specifies the ranking and normalization.

Attribution quality. DT has the lowest RISE on 10 of 13 tasks and the lowest MAS on 11 (Tables 2 and 3). The largest absolute RISE reduction against FT is on MATH, from 0.3487 to 0.2321. DT achieves the highest Recall on all six NIAH tasks and all four VT tasks. At the stated VT budgets, its macro Recall is 87.75%, compared with 76.57% for FT K3. On HotpotQA, DT reaches 85.42% Full-support Recall, compared with 75.00% for FT K3 (Table 4).

Qwen3.5-9B results. Table 5 reports DT and FT results on 13 tasks comprising 520 examples, including all 400 VT and 48 HotpotQA examples. DT has lower RISE on 10 tasks and lower MAS on 12. Recall is higher on all six NIAH and all four VT tasks. On HotpotQA, DT reaches 95.83% Full-support Recall, compared with 58.33% for FT K3, and lowers both RISE and MAS.

Attribution time and memory. The first-page timing panel varies response length on Qwen3-8B. DT, FT, and multi-hop FT (FT-mh) are measured on one MetaX C550. The other curves show the published timing references. Crosses mark out-of-memory runs. On the 16-example Qwen3-8B benchmark, the sum of per-example median attribution times is 11.40 seconds for DT and 12.16 seconds for one-hop FT. Peak allocated memory is 18.76 and 21.49 GB. On Qwen3.5-9B, processing

Table 4: Evidence recovery (% , higher is better) at the listed token budgets. NIAH/VT report token Recall. HotpotQA reports Full-support Recall@30%, the fraction of examples for which every official supporting fact is selected. Sentence scores sum positive token contributions. The budget charges all body tokens. VT macro averages its four rows. FT uses K3 for recovery. Bold marks the highest observed mean.

Task	n	Budget	Perturbation	REAGENT	CLP	IFR	AttnLRP	FT	DT
MQ-Q2	100	10%		39.05	24.35	39.85	47.10	21.49	46.32 67.24
MQ-Q4	100	10%		9.04	8.49	8.58	32.84	20.42	39.58 47.16
MQ-Q8	100	10%		0.97	0.47	0.84	1.19	7.63	7.72 13.59
MV-V2	100	10%		25.48	18.02	20.67	57.50	25.44	53.46 77.79
MV-V4	100	10%		16.05	15.60	14.61	45.24	24.25	49.37 56.79
MV-V8	100	10%		8.02	7.37	7.29	17.87	15.88	19.52 25.19
VT-H2-C3	100	10%		26.69	26.66	29.10	81.16	83.37	72.41 93.81
VT-H4-C1	100	10%		18.86	17.58	20.53	70.09	70.54	68.32 71.31
VT-H6-C1	100	20%		33.57	32.47	33.85	83.74	84.57	82.15 95.37
VT-H10-C1	100	30%		40.70	42.48	44.04	82.66	77.33	83.41 90.51
VT macro	400	mixed		29.96	29.80	31.88	79.41	78.95	76.57 87.75
HotpotQA	48	30%		35.42	35.42	64.58	85.42	87.50	75.00 85.42

Table 5: Attribution quality on Qwen3.5-9B. NIAH uses ten new key/value assignments per task on fixed templates. MATH and MoreHopQA each use six fixed released examples. VT and HotpotQA use all released examples. Recovery is token Recall for NIAH/VT and Full-support Recall@30% for HotpotQA, as defined in Table 4. FT denotes FlashTrace, with K3 for Recall and K1 for RISE/MAS. DT uses signed RISE and positive-part MAS. Bold marks the better mean. Dashes indicate undefined recovery.

Task	n	Budget	Recall (%) $\uparrow$		RISE $\downarrow$		MAS $\downarrow$	
			FT	DT	FT	DT	FT	DT
MQ-Q2	10	10%	62.87	80.59	0.1014	0.1012	0.2535	0.1325
MQ-Q4	10	10%	36.86	52.28	0.1504	0.1585	0.2675	0.2185
MQ-Q8	10	10%	11.97	15.31	0.3393	0.3544	0.4017	0.5044
MV-V2	10	10%	64.30	87.11	0.0971	0.0798	0.2452	0.1095
MV-V4	10	10%	50.41	65.34	0.1068	0.1029	0.2471	0.1390
MV-V8	10	10%	24.27	29.50	0.2357	0.2127	0.3178	0.2965
MATH	6	—	—	—	0.3734	0.2662	0.4801	0.3460
MoreHopQA	6	—	—	—	0.1920	0.1327	0.2763	0.1845
VT H2-C3	100	10%	67.80	94.20	0.1514	0.1095	0.2224	0.1295
VT H4-C1	100	10%	69.07	71.89	0.1161	0.1013	0.1888	0.1343
VT H6-C1	100	20%	83.06	97.35	0.1300	0.1154	0.2051	0.1504
VT H10-C1	100	30%	81.90	94.23	0.1537	0.1561	0.2307	0.2040
HotpotQA	48	30%	58.33	95.83	0.2154	0.1470	0.3125	0.2367

the 16 examples in batches of two with GPU checkpoints reduces time from 16.68 seconds for serial execution to 12.33 seconds, a 35.3% throughput increase. Peak memory is 21.64 and 23.88 GB, respectively. Times cover complete attribution calls. Appendix D gives the measurement protocol.

Reading the token scores. Figures 3 and 4 show retrieval and multi-hop examples with each model’s own scores. Teal marks positive contributions and coral negative contributions to the complete response score. The retrieval example links source keys to their numbers and the query naming them. The multi-hop example follows a voice actor to a film and its director, whose surname determines the answer. Each deletion curve compares DT with one-hop FT on the same model and response. These views connect token-level scores to the evidence selected and the effect of deleting it.

5 RELATED WORK

Reference-based attribution. Integrated Gradients integrates input sensitivities between a baseline and the input, while DeepLIFT propagates activation differences into source contributions (Sundararajan et al., 2017; Shrikumar et al., 2017). Conservative Propagation and AttnLRP develop relevance rules for transformers (Ali et al., 2022; Achibat et al., 2024). DT uses a reference difference for the log-probability of a complete response and supplies finite rules for attention selection and gated memory. The local rules determine how input tokens share that score difference.

Tracing source information. ALTI aggregates token interactions across layers, and Decompx propagates decomposed representations to the prediction head (Ferrando et al., 2022; Modarressi et al., 2023). Information Flow Routes traces influential computations, while FlashTrace aggregates response spans and recursively follows their source importance (Ferrando & Voita, 2024; Pan et al., 2026). DT combines the response positions in one scalar target and follows both the transported values and the controls selecting them.

Recurrent models. MambaLRP conserves relevance through selective state-space computation, and LaTIM decomposes token interactions in Mamba models (Rezaei Jafari et al., 2024; Pitorro & Treviso, 2025). DT follows the retained state, write correction, and query read of Gated Delta Networks (Yang et al., 2025). Its memory and attention paths both track content and control changes, with computation organized through FlashAttention tiles and FLA chunks (Dao et al., 2022; Yang & Zhang, 2024).

6 CONCLUSION

DT assigns signed token contributions to the change in a complete response’s log-probability between reference and original inputs. Its reverse pass propagates coefficients through finite maps, following both the evidence carried by values and the attention or memory operations governing its use. The resulting token scores sum to the response-score difference. Tiled attention and chunked memory make the auxiliary storage linear in sequence length at fixed model dimensions and chunk size. The experiments measure evidence recovery, deletion faithfulness, and attribution cost on attention and hybrid models.

AI USE STATEMENT

AI tools assisted with method development, derivations, coding, experiments, literature review, and manuscript preparation.

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A FORMAL RULES AND DERIVATIONS

A.1 FINITE PROPAGATION AND CONTRIBUTION CONSERVATION

For an operation $v = f(u_1, \dots, u_k)$, DELTATRACE constructs linear maps D_j from the two executions such that

$$\Delta v = \sum_{j=1}^k D_j \Delta u_j. \quad (6)$$

An upstream coefficient m_v propagates to input j as $D_j^\top m_v$. Coefficients from multiple consumers add at their shared input. Starting with coefficient 1 at the scalar target, the reverse traversal produces the source coefficients in Equation 2. For a scalar nonlinearity f , the local coefficient is the divided difference $(f(u_1) - f(u_0))/(u_1 - u_0)$, with its derivative as the coincident-input limit. Linear transformations retain their weights, and addition sends the coefficient to each summand.

Theorem 1 (Contribution conservation). *For an acyclic computation over real-valued operations satisfying Equation 6, reverse propagation from a scalar target yields the conservation identity in Equation 2.*

Proof. An acyclic computation can be expanded into elementary operations with explicit branches. At an operation $v = f(u_1, \dots, u_k)$, Equation 6 gives

$$\langle m_v, \Delta v \rangle = \sum_j \langle D_j^\top m_v, \Delta u_j \rangle. \quad (7)$$

Replacing a node’s contribution by the contributions of its parents therefore preserves the total on a cut through the graph. At shared parents, coefficients from different consumers sum. Repeated substitution reaches the input cut and yields Equation 2, since the coefficient of the scalar target is 1. Constant inputs have zero difference and contribute zero to this sum. $\square$

A.2 FINITE ATTENTION NORMALIZATION

For an attention row $p_b = \text{softmax}(z_b)$, $b \in \{0, 1\}$, the finite rule first converts changes in log-probabilities into changes in probabilities. The coefficient of this conversion is the logarithmic mean ℓ_i . We define it and its sum as

$$\ell_i = \frac{p_{1,i} - p_{0,i}}{\log p_{1,i} - \log p_{0,i}}, \quad \tau = \sum_i \ell_i, \quad (8)$$

with $\ell_i = p_{0,i}$ at coincident probabilities. Dividing by τ gives weights that sum to one for the row averages below. All quantities are evaluated on the common unmasked support.

Proposition 1 (Finite attention normalization). *For finite logits on a common support,*

$$\Delta p = \left(\text{diag}(\ell) - \frac{\ell \ell^\top}{\tau} \right) \Delta z. \quad (9)$$

Proof. The common support contains positive probabilities. For endpoint b , $\log p_b = z_b - a_b \mathbf{1}$, where $a_b = \log \sum_i \exp(z_{b,i})$. The definition of the logarithmic mean gives

$$\Delta p = \ell \odot (\Delta z - \Delta a \mathbf{1}). \quad (10)$$

Probability normalization implies $\mathbf{1}^\top \Delta p = 0$, so

$$\Delta a = \frac{\ell^\top \Delta z}{\tau}. \quad (11)$$

Substitution proves Equation 9. For a target coordinate y ,

$$\Delta \log p_y = \left(e_y - \frac{\ell}{\tau} \right)^\top \Delta z. \quad (12)$$

The coincident-probability definition follows from the continuous limit of the logarithmic mean. $\square$

The operator is symmetric. For an incoming coefficient u_i at probability p_i , the coefficient at its query-key score z_i is

$$m_{z,i} = \ell_i(u_i - \bar{u}_\ell), \quad \bar{u}_\ell = \frac{\sum_j \ell_j u_j}{\sum_j \ell_j}. \quad (13)$$

The response coefficient for a log-softmax target is $m_z = e_y - \ell/\tau$, where e_y is the one-hot vector selecting the target vocabulary coordinate.

A.3 FINITE OPERATOR RULES

For attention output $Y = PV$, Equation 3 gives the coefficients $M_V = \bar{P}^\top M_Y$ and $M_P = M_Y \bar{V}^\top$ for an incoming coefficient matrix M_Y .

Query-key scores use the symmetric product identity,

$$\Delta(QK^\top) = \Delta Q \bar{K}^\top + \bar{Q} \Delta K^\top, \quad \bar{Q} = (Q_0 + Q_1)/2, \quad (14)$$

with $\bar{K}$ defined in the same way. Table 6 lists the finite decompositions by operator. Normalization, rotary position transformations, and grouped heads follow their respective model definitions.

Table 6: Finite propagation rules by operator. Bars denote endpoint averages. The gated-memory recurrence is given in Appendix A.4.

Operation	Change decomposition
Linear and residual	$\Delta(Wx + b) = W\Delta x; \Delta(a + b) = \Delta a + \Delta b$
Attention values	$\Delta(PV) = \bar{P}\Delta V + \Delta P\bar{V}$
Attention scores	$\Delta(QK^\top) = \Delta Q \bar{K}^\top + \bar{Q} \Delta K^\top$
SwiGLU	$\Delta(us) = \bar{s} \odot \Delta u + \bar{u} \odot \Delta s, s = \text{SiLU}(g)$
GDN output gating	$\Delta(n \odot s) = \bar{s} \odot \Delta n + \bar{n} \odot \Delta s$
RMS normalization	Analytic finite rule for $wx/\sqrt{\text{mean}(x^2) + \epsilon}$

A.4 GATED-MEMORY RECURRENCE

With column vectors and a key-by-value state $S_t \in \mathbb{R}^{d_k \times d_v}$, its per-token computation is

$$\begin{aligned} T_t &= \alpha_t S_{t-1}, & r_t &= T_t^\top k_t, \\ c_t &= v_t - r_t, & u_t &= \beta_t c_t, \\ S_t &= T_t + k_t u_t^\top, & o_t &= S_t^\top q_t. \end{aligned} \quad (15)$$

The query includes the model’s fixed attention scale. The state T_t contains retained evidence, r_t reads existing content, and u_t writes the update. One endpoint ordering constructs a finite map D_{GDN}^{01} for the complete memory recurrence through

$$\begin{aligned} \Delta T_t &= \alpha_{t,1} \Delta S_{t-1} + S_{t-1,0} \Delta \alpha_t, \\ \Delta r_t &= (\Delta T_t)^\top k_{t,1} + T_{t,0}^\top \Delta k_t, \\ \Delta u_t &= \beta_{t,1} (\Delta v_t - \Delta r_t) + c_{t,0} \Delta \beta_t, \\ \Delta S_t &= \Delta T_t + k_{t,1} \Delta u_t^\top + \Delta k_t u_{t,0}^\top, \\ \Delta o_t &= (\Delta S_t)^\top q_{t,1} + S_{t,0}^\top \Delta q_t. \end{aligned} \quad (16)$$

Swapping the reference and original activations used as coefficients gives D_{GDN}^{10} . Within each GDN layer, we average these memory-recurrence maps and propagate the readout coefficients as

$$\bar{D}_{\text{GDN}} = \frac{D_{\text{GDN}}^{01} + D_{\text{GDN}}^{10}}{2}, \quad m_{\text{in}} = \bar{D}_{\text{GDN}}^\top m_o. \quad (17)$$

Here m_o collects the coefficients of all readouts o_t in the layer, and m_{in} collects those returned to the recurrence inputs. Equivalently, apply each transpose map to the same m_o and average the returned coefficients. Both maps express the same finite output difference, so their mean preserves it. The recurrence assigns contributions to values, keys, queries, and both gates, carrying credit through stored evidence to earlier sources. Scalar gate nonlinearities and query/key normalization use the surrounding network’s finite rules. Output gating uses the symmetric product in Table 6, with n_t the normalized readout and s_t its activated output-gate vector.

A.5 PRODUCT AND NORMALIZATION IDENTITIES

For a bilinear map B , both

$$\Delta B(a, b) = B(\Delta a, \bar{b}) + B(\bar{a}, \Delta b), \quad (18)$$

$$\Delta B(a, b) = B(\Delta a, b_1) + B(a_0, \Delta b) \quad (19)$$

follow by expansion. Equation 3 and the SwiGLU rule use the first, symmetric identity. The second identity, with the displayed argument ordering, gives the products in Equation 16. Its swapped ordering gives the other layer map averaged in Equation 17. Subtraction and state addition are linear.

For RMS normalization $f(x) = w \odot x/r(x)$, $r(x) = \sqrt{\|x\|^2/d + \epsilon}$, the symmetric product rule gives an analytic coefficient. With $\bar{x} = (x_0 + x_1)/2$, $r_b = r(x_b)$, and $v = w \odot m$, its transpose action is

$$D_f^\top m = \frac{r_0^{-1} + r_1^{-1}}{2} v - \frac{2\bar{x} \langle \bar{x}, v \rangle}{d r_0 r_1 (r_0 + r_1)}. \quad (20)$$

The identity follows from $\Delta r = (2 \langle \bar{x}, \Delta x \rangle / d) / (r_0 + r_1)$ and $\Delta(r^{-1}) = -\Delta r / (r_0 r_1)$. The same construction covers fixed learned scale parameters and the query/key normalization used in the memory path.

B IMPLEMENTATION AND STORAGE ACCOUNTING

Attention. Native endpoint queries, keys, values, and log-normalizers determine the probabilities within each FlashAttention tile. The first sweep accumulates the logarithmic-mean normalizers and weighted centers in Equation 13. Subsequent sweeps accumulate query, key, and value coefficients. Endpoint averages reside in linear-size buffers or are computed inside the tiles.

Memory. The implementation processes 64-token chunks using FLA’s native chunk-state adjoints and batched contractions with saved endpoint states. An associative affine scan computes the finite decay contribution. Convolution and surrounding projections use the installed model primitives and their linear transpose actions.

Storage. For sequence length T , head dimension d , and tile or chunk width C , attention vectors and row statistics occupy $O(Td)$ storage per head. Pairwise products reside in bounded tiles. Dense attention arithmetic is $O(T^2d)$. Chunked memory occupies $O(Td + TC + (T/C)d^2)$ auxiliary storage per head for token coefficients, within-chunk products, and boundary states. Both storage expressions are linear in T at fixed d and C . Layer replay supplies intermediate activations, with checkpoint storage accounted for separately.

C EVALUATION SPECIFICATION

Qwen3-8B uses the released prompts, stored responses, eligible source positions, and evidence annotations. Qwen3.5 NIAH uses new key/value assignments on fixed templates. Table 5 specifies the Qwen3.5 examples for each task. An eligible position is an input position included in the task’s attribution and deletion domain. For NIAH, this domain follows the released token selection. VT and HotpotQA recovery select from the passage body.

Deletion scores. For token attribution scores A , DT ranks tokens by A_i for RISE and by $\max(A_i, 0)$ for MAS. The evaluator records 21 response scores across 20 deletion steps. Each score sums log-probabilities over the complete fixed response plus EOS. Replacing all eligible tokens with EOS gives the attribution reference. Full-input and fully-deleted scores define the normalization endpoints. The normalized values are clipped to $[0, 1]$ and made nonincreasing by a cumulative minimum. For MAS, the remaining attribution share is the positive-score sum over undeleted tokens divided by the initial positive-score sum over eligible tokens. We add its absolute difference from the normalized response score to that score, clip to $[0, 1]$, and rescale the resulting curve to $[0, 1]$ by its minimum and maximum. MAS is the trapezoidal area under this curve. The case-study curves use the positive ranking and share these endpoints between DT and FT for each model and example.

Baseline comparisons. VT and HotpotQA recovery compare all seven methods on the same 448 inputs. The paired development comparison uses the same inputs and stored responses for DT and FT on each model, with one FT recursive step for faithfulness and three for recovery.

Recovery targets and budgets. For NIAH, DT attributes the complete response plus EOS. We select the top 10% of eligible input tokens ranked by positive-part attribution scores. VT scores the answer tokens and ranks positive body-token scores. Its H2/H4/H6/H10 budgets are 10%/10%/20%/30%. HotpotQA scores the full response, ranks passage sentences by the sum of their positive token contributions, and selects the longest ranked sentence prefix within 30% of all body tokens. NIAH/VT report the fraction of annotated evidence tokens recovered. HotpotQA reports Full-support Recall@30%. Each example scores one if every official supporting fact is selected and zero otherwise. The reported value is the mean across examples.

Table 7: Paired development results on the first eight MQ-Q2 and eight MoreHopQA examples per model. These 32 model-example pairs form a separate development set. FT uses one recursive hop for RISE/MAS and three for recovery. RISE is signed, while MAS and recovery use positive scores.

Model	Task	Method	RISE ↓	MAS ↓	Recovery@10% ↑
Qwen3-8B	MQ-Q2	FlashTrace	0.0685	0.1547	56.45
		DELTA TRACE	0.0574	0.0806	77.52
	MoreHopQA	FlashTrace	0.1111	0.1969	—
		DELTA TRACE	0.0585	0.1173	—
Qwen3.5-9B	MQ-Q2	FlashTrace	0.0942	0.2485	63.92
		DELTA TRACE	0.1115	0.1518	73.35
	MoreHopQA	FlashTrace	0.2356	0.2797	—
		DELTA TRACE	0.1688	0.2426	—

D IMPLEMENTATION COST MEASUREMENTS

Each timing comparison uses the same inputs and responses for its alternatives. Complete attribution time covers the two forward executions, the reverse pass, and returning the source scores. Peak allocated device memory includes model weights. Sequence length counts both input and response tokens.

Paired Qwen3 cost. Every method has one warm call followed by three measured calls in rotated order on each of 16 development examples. Per-example medians are aggregated by task. The maximum allocated memory includes model weights. Three-hop FlashTrace takes 18.27 seconds for the same 16 examples.

Tiling on longer released examples. Figure 2 compares dense and tiled finite propagation on the 1,241-token retrieval control, the longest released HotpotQA context (3,470 total tokens), and the longest MATH rollout (3,762 total tokens). At 3,470 tokens, tiling reduces complete attribution time from 5.32 to 3.02 seconds and peak allocation from 32.66 to 22.54 GB. At 3,762 tokens, the corresponding values are 6.12 to 3.40 seconds and 37.18 to 30.06 GB.

Batching and replay retention. The Qwen3.5 comparison processes the same 16 examples serially and in batches of two. With GPU checkpoints and batching, time decreases from 16.68 to 12.33 seconds. Retaining activations for reuse during the reverse pass further reduces time by 5.80% on Qwen3 and 5.15% on Qwen3.5 (Table 8).

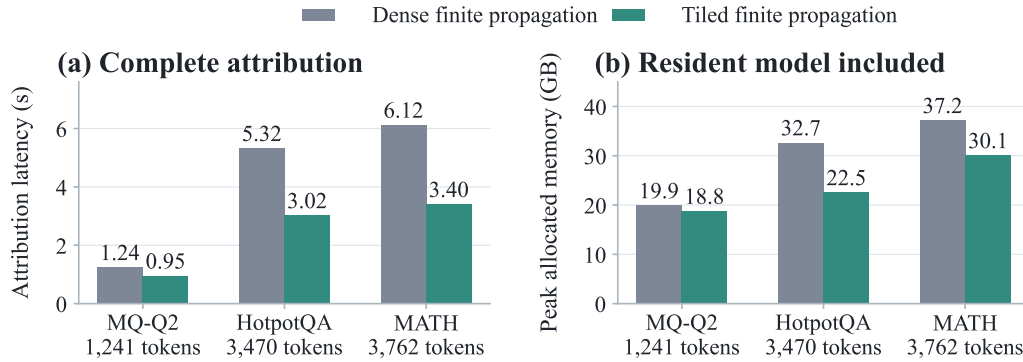


Figure 2: **Dense and tiled finite propagation on three released Qwen3-8B examples.** Complete warm attribution latency is the median of repeated calls. Memory is peak allocated device memory, including resident weights. Each input keeps its original prompt and full stored response. The three groups are distinct tasks and response lengths. Bars show the measured examples.

Table 8: Effect of retaining replay activations on complete attribution cost for 16 examples. Each row compares execution without and with retained replay activations. Warm time is the mean complete pass over 16 examples across two interleaved rounds. Peak memory gives before/after allocations, including resident weights.

Model	Sample batch	Before (s)	Retained (s)	Reduction	Peak GB (before/after)
Qwen3-8B	1	10.360	9.759	5.80%	18.76 / 18.67
Qwen3.5-9B	2	11.480	10.889	5.15%	23.63 / 23.63

E DETAILED SIGNED EVIDENCE VIEWS

(a) Qwen3-8B

Context

One of the special magic
numbers for
deafening-opium is:
5443951.
...
One of the special magic
numbers for mammoth-barber
is: 8698256.

Question

What are all the special
magic numbers for
deafening-opium, and
mammoth-barber mentioned
in the provided text?

Full input

start

end

(b) Qwen3.5-9B

Context

One of the special magic
numbers for
deafening-opium is:
5443951.
...
One of the special magic
numbers for mammoth-barber
is: 8698256.

Question

What are all the special
magic numbers for
deafening-opium, and
mammoth-barber mentioned
in the provided text?

Full input

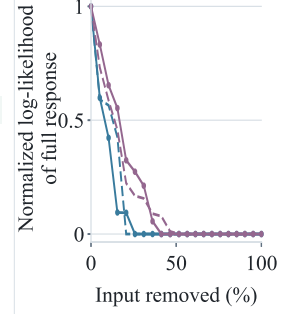
start

end

(c) Deletion

Answer

5443951, 8698256



— Qwen3-8B DT
-- Qwen3-8B FT
— Qwen3.5-9B DT
-- Qwen3.5-9B FT

Signed contribution
(nats)
-9.1 0 +9.1

Figure 3: **Signed evidence for retrieving two numbers.** (a,b) The same `niah_mq-q2` input on Qwen3-8B and Qwen3.5-9B. Teal and coral show positive and negative DT contributions in nats. Marks locate excerpts. (c) Each model’s DT (solid) and one-hop FT (dashed) curves show the normalized log-likelihood of the entire fixed response plus EOS under 20-step deletion. Color identifies the model. The released normalization uses that model’s full-input and fully-deleted scores, clipping and a cumulative minimum. DT deletion uses positive ranking. Answer summarizes the full stored response.

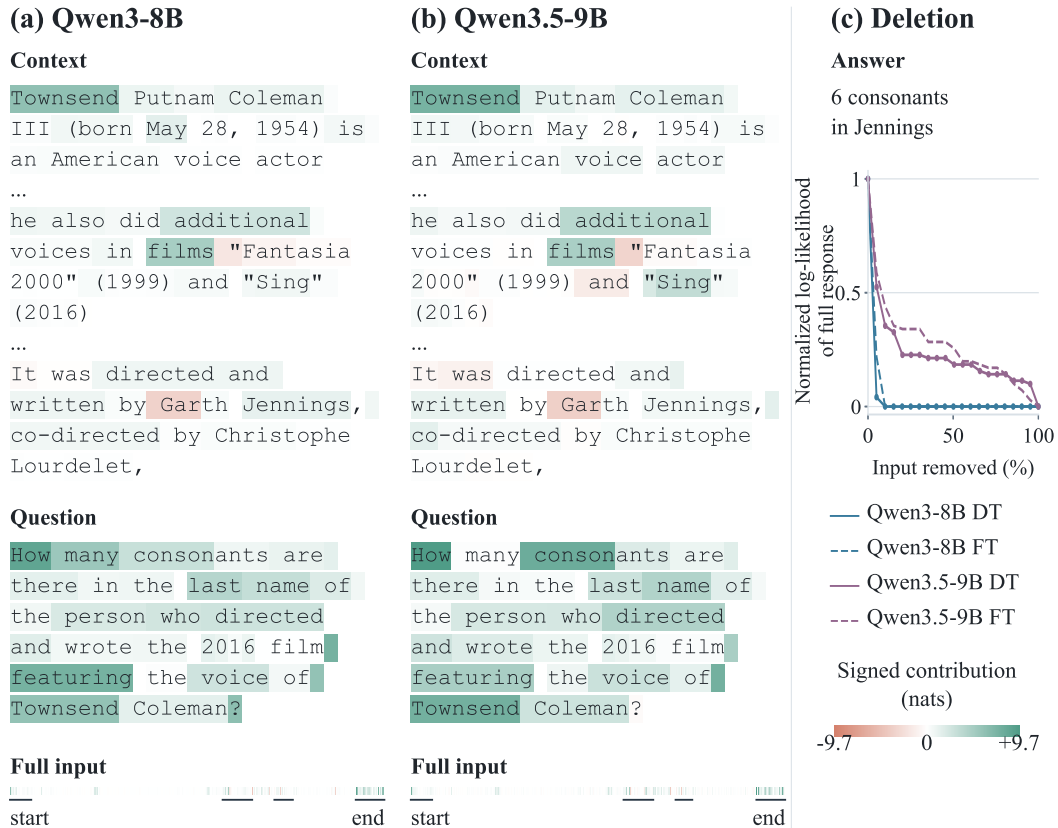


Figure 4: **Signed evidence across a multi-hop question.** (a,b) The MoreHopQA input links Townsend Coleman to *Sing*, Garth Jennings, and his surname’s consonants. (c) Normalized log-likelihood of the entire fixed response plus EOS, comparing each model’s DT with its own one-hop FT baseline. Model colors, method line styles, and per-model normalization follow Figure 3. Both heatmaps share the signed scale within this example.

F COUNTERFACTUAL CREDIT AND POLICY LEARNING

The response attribution studied here suggests a broader research direction based on propagating finite reward differences to actions that contribute to them. The connection to reinforcement learning is particularly direct when counterfactual returns are averaged exactly. The following identity expresses the familiar action-independent-baseline principle (Sutton et al., 1999) in counterfactual form. COMA uses the related construction to assign credit by averaging one agent’s action while fixing the others (Foerster et al., 2018).

Let h be a decision history and π a fixed continuation policy. Write $G^\pi(h, a; \xi)$ for the integrable return obtained by setting the current action to a and subsequently following π , with ξ supplying environment and policy randomness. Define

$$Q^\pi(h, a) = \mathbb{E}_\xi[G^\pi(h, a; \xi)], \quad V^\pi(h) = \mathbb{E}_{a' \sim \pi(\cdot|h)}[Q^\pi(h, a')]. \quad (21)$$

The reference action a' is sampled independently of the selected action a , conditional on h . The two rollouts may share randomness, with each retaining its correct marginal distribution under the corresponding action intervention.

Proposition 2 (Exactly averaged counterfactual credit). *Under these definitions, the policy-averaged counterfactual credit satisfies*

$$C^\pi(h, a) = \mathbb{E}_{a', \xi}[G^\pi(h, a; \xi) - G^\pi(h, a'; \xi)] = Q^\pi(h, a) - V^\pi(h). \quad (22)$$

For a differentiable policy π_θ with positive probabilities over a fixed finite action set, the score-function learning signals obey

$$\mathbb{E}_{a \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a | h) C^{\pi_\theta}(h, a)] = \mathbb{E}_{a \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a | h) Q^{\pi_\theta}(h, a)]. \quad (23)$$

Proof. Linearity of expectation gives Equation 22. The baseline term in Equation 23 is

$$V^{\pi_\theta}(h) \sum_a \pi_\theta(a | h) \nabla_\theta \log \pi_\theta(a | h) = V^{\pi_\theta}(h) \nabla_\theta \sum_a \pi_\theta(a | h) = 0. \quad (24)$$

Subtracting this term proves the result. Credit and action value serve as weights in the score-function update. $\square$

This equality provides a concrete target for extending finite attribution to learning. The averaged action credit of a reward-level construction should recover the advantage.